Basic interview questions

1) React:

React is an open-source JS library developed by facebook for building user interfaces, especially for single-page applications. It allows developers to create reusable UI components and efficiently update the DOM using a virtual DOM.

2) React framework or library:

React is a library, not a framework. It only handles the view layer (UI rendering). Unlike frameworks (e.g Angular), React doesn't include built-in solutions for routing, state management, or HTTP calls - you use third-party tools for those.

3) Key features:

Virtual DOM, component based architecture, JSX syntax, unidirectional data flows, Hooks (useState, useEffect, etc)., and Reusability.

4) SPA : (Single Page Application):

An SPA is a web app that loads a single HTML page and dynamically updates content as the user interacts, without full page reloads. React is commonly used to build SPAs.

5) Components in React :

A component is an independent, reusable piece of UI. React apps are built by composing multiple components together. Each component can have its own state & logic

6) Functional components:

Functional components are plain JS functions that return JSX. Since React 16.8, they support Hooks like (useState, useEffect), making them the preferred way to write components.

JSX:

JavaScript (XML) is a syntax extension for JavaScript that lets you write HTML-like code inside JavaScript. It makes the UI structure more readable and is transpiled to React.createElement() calls by Babel.

Browsers understand JSX directly?

No. Browsers only understand plain JavaScript. JSX must be transpiled into regular JavaScript using a tool like Babel before the Browser can execute it.

Babel in React:

Babel is a JS transpiler (compiler) that converts modern JS & JSX into backwards-compatible JavaScript that browsers can understand. In React projects, it converts JSX like Hello into React.createElement('div', null, 'Hello').

(c) Virtual DOM:

The virtual DOM is a lightweight, in-memory copy of the real DOM. When state or props change, React first updates the virtual DOM, then compares it with the previous version (called diffing), and only applies the minimal necessary changes to the real DOM - making updates faster and more efficient.